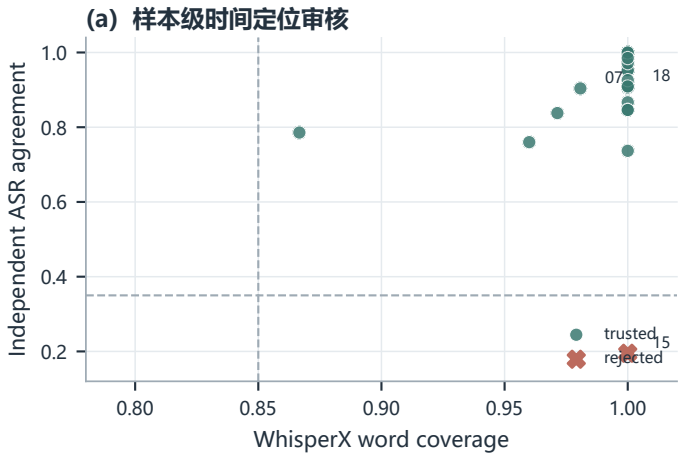
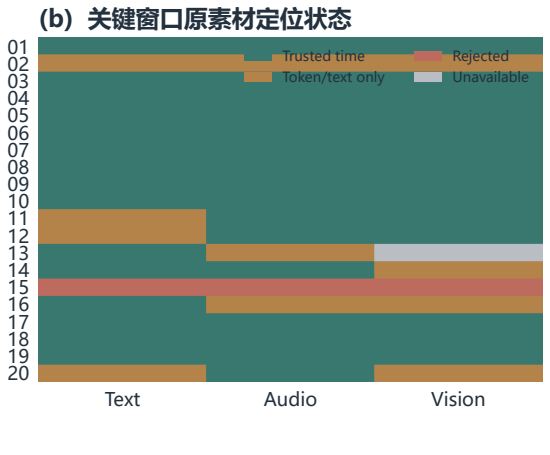
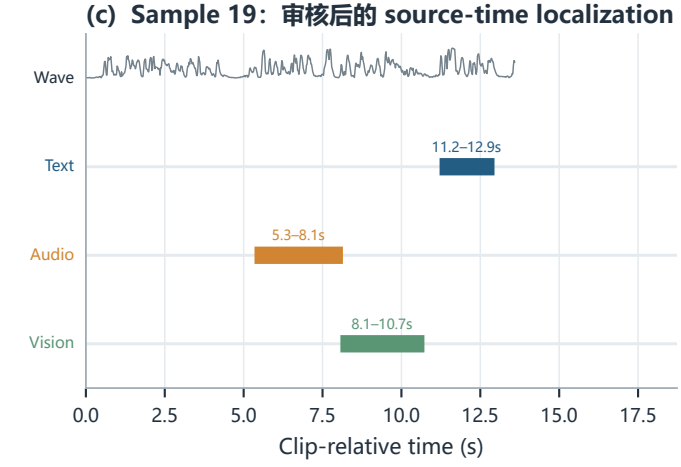

(a) 样本级时间定位审核
Independent ASR agreement
1.0
0.8
0.6
0.4
0.2
0.80
0.85
0.90
0.95
1.00
WhisperX word coverage
07
18
15
trusted
rejected


(b) 关键窗口原素材定位状态
01
02
03
04
05
06
07
08
09
10
11
12
13
14
15
16
17
18
19
20
Trusted time
Rejected
Token/text only
Unavailable
Text
Audio
Vision


(c) Sample 19：审核后的 source-time localization
Wave
Text
Audio
Vision
11.2–12.9s
5.3–8.1s
8.1–10.7s
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Clip-relative time (s)


(d) Sample 19 原视频帧 • 9.40s